1. A planet has double the mass of the earth. Its average density is equal to that of the earth. An object weighing W on earth will weigh on that planet:
[2023 (06 Apr Shift 1)]
(1) $2^{\frac{1}{4}}\mathrm{W}$
(2) $2^{\frac{1}{3}}\mathrm{W}$
(3) 2W
(4) $2^{\frac{2}{3}}\mathrm{W}$

2. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Earth has atmosphere whereas moon doesn't have any atmosphere.
Reason R: The escape velocity on moon is very small as compared to that on earth.
In the light of the above statements, choose the correct answer from the options given below:
[2023 (06 Apr Shift 1)]
(1) Both A and R are correct but R is NOT the correct explanation of A
(2) A is false but R is true
(3) Both A and R are correct and R is the correct explanation of A
(4) A is true but R is false

3. The weight of a body on the surface of the earth is 100 N. The gravitational force on it when taken at a height, from the surface of earth, equal to one-fourth the radius of the earth is:
[2023 (06 Apr Shift 2)]
(1) 64 N
(2) 25 N
(3) 50 N
(4) 100 N

4. Choose the incorrect statement from the following:
[2023 (06 Apr Shift 2)]
(1) The speed of satellite in a given circular orbit remains constant
(2) For a planet revolving around the sun in an elliptical orbit, the total energy of the planet remains constant
(3) The linear speed of a planet revolving around the sun remains constant
(4) When a body falls towards earth, the displacement of earth towards the body is negligible

5. Given below are two statements:
Statement I: If E be the total energy of a satellite moving around the earth, then its potential energy will be $\frac{E}{2}$.
Statement II: The kinetic energy of a satellite revolving in an orbit is equal to the half the magnitude of total energy E.
In the light of the above statements, choose the most appropriate answer from the options given below.
[2023 (08 Apr Shift 1)]
(1) Statement I is correct but Statement II is incorrect
(2) Statement I is incorrect but Statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

6. The weight of a body on the earth is 400 N. Then weight of the body when taken to a depth half of the radius of the earth will be:
[2023 (08 Apr Shift 1)]
(1) 200 N
(2) Zero
(3) 100 N
(4) 300 N

7. The orbital angular momentum of a satellite is L, when it is revolving in a circular orbit at height h from earth surface. If the distance of satellite from the earth centre is increased by eight times to its initial value, then the new angular momentum will be
[2023 (08 Apr Shift 2)]
(1) 8 L
(2) 9 L
(3) 4 L
(4) 3 L

8. The acceleration due to gravity at height h above the earth if $h \ll R$ (Radius of earth) is given by
[2023 (08 Apr Shift 2)]
(1) $g' = g\left(1 - \frac{h^2}{2R^2}\right)$
(2) $g' = \mathrm{g}\left(1 - \frac{\mathrm{h}}{2\mathrm{R}}\right)$
(3) $g' = \mathrm{g}\left(1 - \frac{2\mathrm{h}}{\mathrm{R}}\right)$
(4) $g' = g\left(1 - \frac{2h^2}{R^2}\right)$

9. Two satellites of masses m and 3 m revolve around the earth in circular orbits of radii r & 3r respectively. The ratio of orbital speeds of the satellites respectively is
[2023 (10 Apr Shift 1)]
(1) $\sqrt{3} : 1$
(2) $3 : 1$
(3) $9 : 1$
(4) $1 : 1$

10. Assuming the earth to be a sphere of uniform mass density, the weight of a body at a depth $d = \frac{R}{2}$ from the surface of earth, if its weight on the surface of earth is 200 N, will be : (Given $R =$ radius of earth)
[2023 (10 Apr Shift 1)]
(1) 300 N
(2) 100 N
(3) 400 N
(4) 500 N

11. If the earth suddenly shrinks to $\frac{1}{64}$th of its original volume with its mass remaining the same, the period of rotation of earth becomes $\frac{24}{x}h$. The value of x is ______
[2023 (10 Apr Shift 1)]

12. Given below are two statements:
Statement I : Rotation of the earth shows effect on the value of acceleration due to gravity (g).
Statement II : The effect of rotation of the earth on the value of g at the equator is minimum and that at the pole is maximum.
In the light of the above statements, choose the correct answer from the options given below
[2023 (10 Apr Shift 2)]
(1) Statement I is false but statement II is true
(2) Both Statement I and Statement II are true
(3) Both Statement I and Statement II are false
(4) Statement I is true but statement II is false

13. The time period of a satellite, revolving above earth's surface at a height equal to R will be (Given $g = \pi^2 \mathrm{~m~s}^{-2}$, $R =$ radius of earth)
[2023 (10 Apr Shift 2)]
(1) $\sqrt{2\mathrm{R}}$
(2) $\sqrt{8\mathrm{R}}$
(3) $\sqrt{32\mathrm{R}}$
(4) $\sqrt{4\mathrm{R}}$

14. The radii of two planets A and B are R and $4R$ and their densities are ρ and $\frac{\rho}{3}$ respectively. The ratio of acceleration due to gravity at their surfaces $(g_A : g_B)$ will be
[2023 (11 Apr Shift 1)]
(1) $4 : 3$
(2) $1 : 16$
(3) $3 : 16$
(4) $3 : 4$

15. A space ship of mass 2×10^4 kg is launched into a circular orbit close to the earth surface. The additional velocity to be imparted to the space ship in the orbit to overcome the gravitational pull will be (if $g = 10 \mathrm{~m~s}^{-2}$ and radius of earth $= 6400$ km):
[2023 (11 Apr Shift 2)]
(1) $11.2\left(\sqrt{2} - 1\right) \mathrm{~km~s}^{-1}$
(2) $8\left(\sqrt{2} - 1\right) \mathrm{~km~s}^{-1}$
(3) $7.9\left(\sqrt{2} - 1\right) \mathrm{~km~s}^{-1}$
(4) $7.4\left(\sqrt{2} - 1\right) \mathrm{~km~s}^{-1}$

16. Two satellites A and B move round the earth in the same orbit. The mass of A is twice the mass of B. The quantity which is same for the two satellites will be

[2023 (12 Apr Shift 1)]

(1) Speed

(2) Kinetic energy

(3) Total energy

(4) Potential energy

17. The ratio of escape velocity of a planet to the escape velocity of earth will be:-

Given: Mass of the planet is 16 times mass of earth and radius of the planet is 4 times the radius of earth.

[2023 (12 Apr Shift 1)]

(1) $4 : 1$

(2) $1 : 4$

(3) $1 : \sqrt{2}$

(4) $2 : 1$

18. A planet having mass 9 M_e and radius $4R_e$, where M_e and R_e are mass and radius of earth respectively, has escape velocity in $\mathrm{km\ s^{-1}}$ given by: (Given escape velocity on earth $V_e = 11.2 \times 10^3\ \mathrm{m\,s^{-1}}$)

[2023 (13 Apr Shift 1)]

(1) 67.2

(2) 16.8

(3) 11.2

(4) 33.6

19. Two planets A and B of radii R and $1.5\ R$ have densities ρ and $\frac{\rho}{2}$ respectively. The ratio of acceleration due to gravity at the surface of B to A is:

[2023 (13 Apr Shift 2)]

(1) $2 : 3$

(2) $2 : 1$

(3) $3 : 4$

(4) $4 : 3$

20. Given below are two statements:

Statement I : For a planet, if the ratio of mass of the planet to its radius increase, the escape velocity from the planet also increase.

Statement II : Escape velocity is independent of the radius of the planet.

In the light of above statements, choose the **most appropriate** answer from the options given below

[2023 (13 Apr Shift 2)]

(1) Statement I is incorrect but Statement II is correct

(2) Statement I is correct but statement II is incorrect

(3) Both Statement I and Statement II are incorrect

(4) Both Statement I and Statement II are correct

21. Two identical particles each of mass m go round a circle of radius a under the action of their mutual gravitational attraction. The angular speed of each particle will be :

[2023 (15 Apr Shift 1)]

(1) $\sqrt{\frac{Gm}{a^3}}$

(2) $\sqrt{\frac{Gm}{8a^3}}$

(3) $\sqrt{\frac{Gm}{4a^3}}$

(4) $\sqrt{\frac{Gm}{2a^3}}$

ANSWER KEYS

1. (2)	**2.** (3)	**3.** (1)	**4.** (3)	**5.** (4)	**6.** (1)	**7.** (4)	**8.** (3)
9. (1)	**10.** (2)	**11.** (16)	**12.** (4)	**13.** (3)	**14.** (4)	**15.** (2)	**16.** (1)
17. (4)	**18.** (2)	**19.** (3)	**20.** (2)	**21.** (3)			

1. (2)

The formula for acceleration due to gravity is

$g = \frac{GM}{R^2}$

Let the density of earth and planet be ρ, and the radius of the planet be R'. Let R be the radius of the earth and g be the acceleration due to gravity of earth and g' be the acceleration due to gravity of planet.

So, the masses are

$M = \frac{4}{3}\pi R^3 \rho \quad \ldots (i)$

$M' = \frac{4}{3}\pi (R')^3 \rho \ldots (ii)$

So, $\frac{g'}{g} = \frac{\frac{GM'}{R'^2}}{\frac{GM}{R^2}} = \frac{M'}{M}\left(\frac{R}{R'}\right)^2 \quad \ldots (iii)$

From equation (i) and (ii) we get $R' = 2^{\frac{1}{3}}R$

Substituting the value in equation (iii),

$\frac{g'}{g} = (2^1)2^{\frac{-2}{3}} = 2^{\frac{1}{3}}g$

Weight is given by $W' = mg' = 2^{\frac{1}{3}}W$

2. (3)

The formula to calculate the escape velocity is given by

$v_e = \sqrt{2gR} \quad \ldots (1)$

As both the acceleration and radius of the Moon is much less than that of the Earth, the value of the escape velocity in Moon is much less than that on the Earth.

The average velocity of gas molecules remains higher than the escape velocity due to the lower escape velocity on the moon's surface, making the moon unable to maintain an atmosphere.

3. (1)

The acceleration due to gravity changes as one goes to a height from the surface of the earth. Let the new acceleration due to gravity be g'. The formula for acceleration due to gravity at a height is given by $g' = \frac{gR^2}{(R+h)^2}$

The weight of the body on surface is $W = mg = 100$ N

The weight at the given height is $W = mg$'.

$\Rightarrow W' = 100 \times \frac{R^2}{\left(R+\frac{R}{4}\right)^2}$

$\Rightarrow W' = 100 \times \frac{R^2}{\frac{25R^2}{16}} = 64$ N

4. (3)

A planet revolves round the Sun by the action of the gravitational force between the sun and the planet. Depending up on the distance from the sun in the elliptic orbit, the speed of revolution of the planet changes.

The formula to calculate the linear speed of a planet in an elliptical orbit can be written as

$v = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)}$

From the above expression, it can be concluded that the linear speed of the planet is not constant during its revolution.

5. (4)

Let m be the mass of the satellite.
The potential energy of the satellite is
$U = \frac{-GMm}{r}$
The velocity of the satellite is given by
$v = \sqrt{\frac{GM}{r}}$
Kinetic energy is given by
$K = \frac{mv^2}{2} = \frac{m}{2}\frac{GM}{r} = \frac{GMm}{2r}$
The total mechanical energy is the sum of potential and kinetic energy. So,
$T = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}$
Hence, both the statements are false.

6. (1)

The acceleration due to gravity changes with depth from the surface, which is given by
$g' = g\left(1 - \frac{\mathrm{d}}{\mathrm{R}}\right) \quad \ldots\left(\mathrm{i}\right)$
The given data is
$W = 400\ \mathrm{N}$
$\mathrm{d} = \frac{\mathrm{R}}{2}$
Multiplying equation (i) with the mass we get the weight.
$mg' = mg\left(1 - \frac{\frac{R}{2}}{R}\right) = mg\left(\frac{1}{2}\right)$
$\Rightarrow W' = \frac{400\ \mathrm{N}}{2} = 200\ \mathrm{N}$

7. (4)

For a satellite revolving around the Earth in a circular orbit, it can be written that
$\frac{mv^2}{r} = \frac{GMm}{r^2} \quad \ldots(1)$ ($r = R + h$ is the distance from the centre of the Earth)
Simplify equation (1) to obtain the linear speed of the satellite.
$v = \sqrt{\frac{GM}{r}}$
Thus, the angular momentum of the satellite is given by
$L = mvr$
$= mr\sqrt{\frac{GM}{r}}$
$= m\sqrt{GMr} \quad \ldots(2)$
Similarly, for distance r' from the centre of the Earth, the angular momentum is given by
$L' = m\sqrt{GMr'} \quad \ldots(3)$
Divide equation (3) by equation (2) to obtain the required angular momentum.
$\frac{L'}{L} = \frac{m\sqrt{GMr'}}{m\sqrt{GMr}}$
$= \sqrt{\frac{r'}{r}} \quad \ldots(4)$
Now, $r' = r + 8r = 9r$
Substitute the known values of the parameters into equation (4) to obtain the required angular momentum.
$\frac{L'}{L} = \sqrt{9}$
$\Rightarrow L' = 3L$

8. (3)

The acceleration due to gravity g' at a height h from the surface of Earth is given by

$g' = \frac{GM}{(R+h)^2} \quad \ldots(1)$

Use the method of binomial expansion to obtain the approximate expression for the acceleration due to gravity at a particular height from the surface of the Earth.

$$g' = \frac{GM}{R^2\left(1+\frac{h}{R}\right)^2}$$

$$= \frac{GM}{R^2}\left(1+\frac{h}{R}\right)^{-2}$$

$$\approx \frac{GM}{R^2}\left(1-\frac{2h}{R}\right) \quad \ldots(2)$$

The formula to calculate the acceleration due to gravity at the surface of the Earth is given by

$g = \frac{GM}{R^2} \quad \ldots\left(3\right)$

Divide equation (2) by equation (3) and simplify to obtain the required relation.

$$\frac{g'}{g} = \frac{\frac{GM}{R^2}\left(1-\frac{2h}{R}\right)}{\frac{GM}{R^2}}$$

$$= \left(1-\frac{2h}{R}\right)$$

$$\Rightarrow g' = g\left(1-\frac{2h}{R}\right)$$

9. (1)

The orbital velocity is given

$V = \sqrt{\left(\frac{GM}{r}\right)}$

where M is the mass of earth and r is the radius of the orbit of satellite.

The velocity of the satellite with mass m is

$V_1 = \sqrt{\frac{GM}{r}}$

and the velocity of the satellite with mass $3m$ is

$V_2 = \sqrt{\frac{GM}{3r}}$

Hence, the ratio of the velocities is

$\frac{V_1}{V_2} = \sqrt{\frac{3r}{r}} = \frac{\sqrt{3}}{1}$

10. (2)

The acceleration due to gravity with depth is

$g' = g\left(1-\frac{d}{R}\right)$

The given data is

$W = 200\ \mathrm{N}$

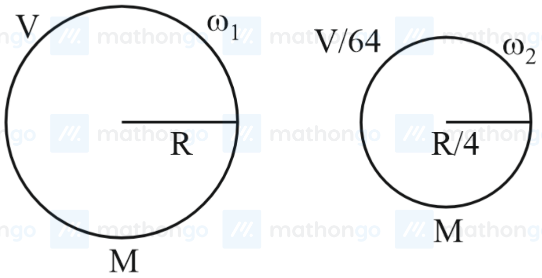
$d = \frac{\mathrm{R}}{2}$

Let the weight be at the given depth, W'. Hence, the value is

$$mg' = mg\left(1-\frac{\frac{R}{2}}{R}\right) = m\left(\frac{g}{2}\right)$$

The Weight $W' = \frac{W}{2} = \frac{200}{2} = 100\ \mathrm{N}$

11. (16)



$L =$ constant; $L = I\omega$

From the conservation of angular momentum,

$\frac{2}{5}MR^2\omega_1 = \frac{2}{5}M\left(\frac{R}{4}\right)^2\omega_2$

$\Rightarrow MR^2\omega_1 = \frac{MR^2\omega_2}{16}$

$\Rightarrow \frac{\omega_1}{\omega_2} = \frac{1}{16}$

Using $\omega = \frac{2\pi}{T}$ for each case along with the respective angular frequencies,

$\frac{T_2}{T_1} = \frac{1}{16}$

$\Rightarrow \frac{T_1}{T_2} = \frac{16}{1}$

Using $T_1 = 24$ h, $x = 16$ by comparing with the given time period in the question.

12. (4)

The variation of acceleration due to gravity with the rotation of Earth can be written as

$g' = g - \omega^2 R\cos^2\lambda \quad \ldots(1)$

Substitute 0° for λ for equatorial position to obtain the acceleration due to gravity.

$g' = g - \omega^2 R\cos^2 0^\circ$

$= g - \omega^2 R$

Substitute 90° for λ for polar position to obtain the acceleration due to gravity.

$g' = g - \omega^2 R\cos^2 90^\circ$

$= g$

The above two equations show that the effect on acceleration due to gravity is maximum at the equator and minimum at pole.

13. (3)

The velocity of the satellite revolving round the Earth at a height r from the centre of the Earth is given by

$v = \sqrt{\left(\frac{GM}{r}\right)} \quad \ldots(1)$

The formula to calculate the time period in the circular orbit is given by

$T = \frac{2\pi r}{v} \quad \ldots(2)$

In this case: $r = R + R = 2R$

Substitute the expression for the velocity from equation (1) into equation (2) and simplify to obtain the required time period.

$\mathrm{T} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$

$= \frac{2\pi\times 2R}{\sqrt{\frac{gR}{2}}}$

$= \frac{4\pi R\sqrt{2}}{\pi\sqrt{R}}$

$= 4\sqrt{2R}$

$= \sqrt{32R}$

14. (4)

Acceleration due to gravity is written as

$g=\frac{GM}{r^2}$

$=\frac{G\times\frac{4}{3}\times\pi\times r^3\times\rho}{r^2}$

$=\left(\frac{4\pi G}{3}\right)\rho r$

Given:

$r_A=R$
$r_B=4R$
$\rho_A=\rho$
$\rho_B=\frac{\rho}{3}$

Hence, the ratio is

$\frac{g_A}{g_B}=\frac{R\times\rho}{4R\times\frac{\rho}{3}}=\frac{3}{4}.$

15. (2)

The orbital velocity of satellite is

$v_{orb}=\sqrt{\frac{GM}{R}}$

The escape velocity is

$v_{esc}=\sqrt{\frac{2GM}{R}}$

Hence, the additional velocity required is

$\Delta v=v_{esc}-v_{orb}=\sqrt{\frac{2GM}{R}}-\sqrt{\frac{GM}{R}}=\left(\sqrt{2}-1\right)\sqrt{\frac{GM}{R}}$

$=\left(\sqrt{2}-1\right)\sqrt{gR}$

$=\left(\sqrt{2}-1\right)\sqrt{10\times6400\times10^3}$

$=8\left(\sqrt{2}-1\right)\ \text{km s}^{-1}$

16. (1)

The formula to calculate the speed of a satellite while revolving around the Earth is given by

$v=\sqrt{\frac{GM_e}{R}}\quad\ldots(1)$

where, v is the speed, G is the Gravitational constant, M_e is the mass of the Earth, R is the radius of the orbit.

As can be seen from the equation, speed is independent of mass of satellite.

Hence, the speed of the satellites will be the same.

17. (4)

The formula to calculate the escape velocity is given by

$v_e=\sqrt{\frac{2GM}{R}}\quad\ldots(1)$

For the Earth, it is given by

${v_e}^E=\sqrt{\frac{2GM_E}{R_E}}\ldots(2)$

and, for the planet, it is given by

${v_e}^P=\sqrt{\frac{2GM_P}{R_P}}\ldots(3)$

Divide equation (2) by equation (3) and simplify to obtain the required ratio.

$\frac{{v_e}^E}{{v_e}^P}=\sqrt{\frac{M_E}{M_P}}\times\sqrt{\frac{R_P}{R_E}}\quad\ldots(4)$

Substitute the values of the known parameters into equation (4) to obtain the required ratio.

$\frac{{v_e}^E}{{v_e}^P}=\sqrt{\frac{M_E}{16M_E}}\times\sqrt{\frac{4R_E}{R_E}}$

$=\frac{1}{2}$

$\Rightarrow {v_e}^P:{v_e}^E=\ 2:1$

18. (2)

The escape velocity on earth is given by $V_e = \sqrt{\frac{2GM_e}{R_e}}$

Escape velocity for the planet is

$V'_{esc} = \sqrt{\frac{2G(9M_e)}{4R_e}} \quad \ldots \left(i\right)$

The escape velocity of earth is $V_e = 11.2 \times 10^3 \text{ m s}^{-1}$

From equation (i)

$V'_{esc} = \sqrt{\frac{9}{4}\left(\frac{2GM_e}{R_e}\right)} = \frac{3}{2} \times 11.2 \times 10^3 \text{ m s}^{-1}$

$\Rightarrow \text{V'}_{\text{esc}} = 16.8 \text{ km s}^{-1}$

19. (3)

The acceleration due to gravity is

$g = \frac{GM}{R^2} = \frac{4\pi GR\rho}{3}$

$\Rightarrow g \propto \rho \cdot R$

Thus, the ratio is

$\Rightarrow \frac{g_B}{g_A} = \frac{1.5R\rho}{2\rho R} = \frac{1}{2} \times 1.5 = 0.75$

$\Rightarrow \frac{g_B}{g_A} = \frac{3}{4}$

20. (2)

The escape velocity is given by the formula

$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$

Clearly, the escape velocity depends on the mass and radius of the planet. Hence, if the ratio increases the escape velocity also increases.

So, Statement **I** is correct while statement **II** is wrong.

21. (3)

The gravitational force (F_G) between the particles is given by

$F_G = \frac{Gm^2}{(2a)^2}$

$= \frac{Gm^2}{4a^2} \quad \ldots(1)$

The centripetal force (F_C) of each particle can be written as

$F_C = m\omega^2 a \quad \ldots(2)$

Under balanced condition, equate equation (1) and equation (2) and simplify to obtain the angular speed of each particle.

$m\omega^2 a = \frac{Gm^2}{4a^2}$

$\Rightarrow \omega^2 = \frac{Gm}{4a^3}$

$\Rightarrow \omega = \sqrt{\frac{Gm}{4a^3}}$